LESSON PLAN · WEB DEVELOPMENT (GR 6–8) BEGINNER: WEB CREATORS · LEVEL 1

Lesson 7: First JavaScript

Module 2 | 60 minutes | script, events, variables | Grades 6–11 · ages 11–16

Objective: Use JavaScript to make a button change text and colour, and keep count with a variable.

Success criteria

- I can add JavaScript in a script tag.
- I can find an element with getElementById.
- I can make a button react with addEventListener.
- I can keep score with a variable.

Key vocabulary

- **JavaScript**: The language that makes web pages react and move.
- **script tag**: The tag that holds JavaScript code.
- **id**: A unique name for one element, used to find it.
- **event**: Something that happens, like a click.
- **event listener**: Code that waits for an event and then runs.
- **variable**: A named box that stores a value, made with let.

You will need

- Projector with Kitabash (lesson video, Web Lab)
- Devices: 1 per learner, with a keyboard
- Printed worksheet (optional)
- Project files: l07-click-counter.html, l07-teacher-demo.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game: Human Event Listener**	Play Human Event Listener with ids and claps; a scorekeeper adds up on the whiteboard.
5 min	**Hook and video**	Show the hook. Thumbs poll: “Have you played a game in a web browser?” Play the video; pause at “Test it in the live preview” and ask what to check if nothing happens.
10 min	**Live demo and build-along**	Run the demo in the Web Lab, including the id typo hunt. Everyone types the click counter, clicks it in the preview and presses Check.
8 min	**Challenge and games**	Web Lab: make the button talk, then Predict the score, the match game and Order it. Quick competition: first correct score prediction shouts “Howzat!”.

17 min	**Project in pairs**	Breakout rooms: pairs build a score counter or click-the-mango game. The navigator clicks every button in the preview to test it. Swap roles every 5 minutes.
5 min	**Probe: debug race**	Race to fix “the silent scoreboard”. First learner to explain both bugs wins.
6 min	**Show-and-tell: play my game**	Two learners share their browser and play their game live; the class calls out what to click.
4 min	**Wrap-up and exit ticket**	Fist-to-five on the success criteria. Exit ticket: add 4 to runs.

Unplugged warm-up

Human Event Listener (5 min game): each learner is given an “id” (bat, ball, stumps). The teacher calls “getElementById bat — on click, jump!” Then the teacher claps (the click) and only “bat” must jump. Add a scorekeeper who adds 1 to a running total on the whiteboard each time. Anyone who moves on the wrong id is out.

Teaching the idea

JavaScript makes a page react. Code goes in a script tag at the bottom of the page. document.getElementById finds an element by its id; addEventListener("click", ...) runs code each time it is clicked; textContent changes the words; style changes CSS. A variable made with let stores a value, and score = score + 1 updates it. The Web Lab live preview runs JavaScript in a locked-down sandbox, so learners click their buttons right in the preview. Saving the page as an .html file to play at home is an optional extra.

Analogy: An event listener is like a fielder waiting at slip: they do nothing until the ball comes (the click), then they spring into action every single time.

Live demo: 1) Open the Web Lab and explain: the live preview runs JavaScript, so today we click our buttons right in the preview. 2) Type a paragraph with id="count" and a button with id="btn". Click the button in the preview: nothing happens yet. 3) Add a <script> tag at the bottom. Type const btn = document.getElementById("btn"); and explain it finds the button by its id. 4) Type btn.addEventListener("click", function () { ... }); and inside it count.textContent = "Clicked!"; Click the button in the preview and celebrate. 5) Add let clicks = 0; and clicks = clicks + 1; then show "Clicks: " + clicks. Click five times. 6) Misspell the id as "bnt" and click again: nothing happens. Ask learners to hunt for the typo by comparing the id in the HTML with the one in getElementById. Fix it together. Optional: show the finished page saved as an .html file and opened in Chrome or Edge, so learners know they can play it at home.

Questions to ask

Ask	Listen for
Where does the script tag usually go?	At the bottom of the page, after the HTML it uses.

How does JavaScript find the button?	document.getElementById with the button's id.
What does addEventListener do?	Waits for an event like a click, then runs the code inside.
What does score = score + 1 do?	Takes the old score, adds one and stores it back.
My button does nothing when I click it. What should I check first?	That the id in getElementById matches the id in the HTML exactly, and that every quote has a partner.

Differentiation

Support
Give learners the build-along counter to change: new button words, a new message, +2 instead of +1. Provide a printed "find, listen, change" pattern card.

Stretch
Add a reset button, a target score that shows "You win!" using if, and change the button colour on each click.

Common misconceptions

Misconception	How to fix it
getelementbyid works the same as getElementById	JavaScript is case-sensitive. Capitals must be exact: getElementById.
The script runs again by itself	Only code inside the click function runs again on each click.
textContent changes the colour	textContent changes words; use style to change colours.
Text does not need quotes in JavaScript, like textContent = Hello	Words must go inside quotes: textContent = "Hello". Without quotes JavaScript thinks Hello is a variable name.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write one line of JavaScript that adds 4 to a variable called runs.

Homework
Add one new button or a reset button to your game and test it in the preview. Optional: save your code as an .html file and play it with someone at home in any browser, then teach them what addEventListener means.

Answer key — Kitabash activities

1. Web Lab: make the button talk (perform · web lab)

Goals checked: Has a script · Finds elements by id · Listens for a click · Changes the words

One solution:

```
<h1>Secret message</h1>
<p id="message">Press the button to reveal the secret.</p>
<button id="reveal">Reveal</button>

<script>
  const reveal = document.getElementById("reveal");
  const message = document.getElementById("message");
  reveal.addEventListener("click", function () {
    message.textContent = "The secret is: coding is a superpower!";
  });
</script>
```

2. Predict the score (perform · code output)

3. Match each piece of JavaScript to its job (perform · matching)

<script> → Holds the JavaScript code · let score = 0; → Makes a variable that starts at zero · document.getElementById("btn") → Finds the element with id="btn" · addEventListener("click", ...) → Waits for a click, then runs code · textContent → The words inside an element · score = score + 1; → Adds one to the score

4. Order it: build a counter (perform · ordering)

Order: Add a button and a paragraph, each with an id → Add a script tag at the bottom of the page → Make a variable: let clicks = 0 → Find the button and paragraph with getElementById → Listen for a click and add 1 to clicks → Show the new number with textContent

5. Project: score counter or click-the-mango game (perform · web lab)

Goals checked: Has at least two buttons · Has a script · Keeps score in a variable · Reacts to clicks and shows the score · Is styled

Example page:

```
<!-- Cricket score counter -->
<style>
  body {
    font-family: Arial, sans-serif;
    text-align: center;
    background-color: #e8f6ef;
  }
  #score {
    font-size: 48px;
    color: #1e8449;
  }
  button {
    font-size: 20px;
    padding: 12px 18px;
    margin: 6px;
    border-radius: 10px;
    border: 2px solid #1e8449;
    background-color: white;
  }
</style>

<h1>🏏 Cricket Score Counter</h1>
<p id="score">0 / 0</p>
<p id="message">Let’s play!</p>
<button id="one">+1 run</button>
<button id="four">FOUR!</button>
<button id="six">SIX!</button>
<button id="out">Wicket</button>

<script>
  let runs = 0;
  let wickets = 0;
  const score = document.getElementById("score");
  const message = document.getElementById("message");

  function show(text) {
    score.textContent = runs + " / " + wickets;
    message.textContent = text;
  }

  document.getElementById("one").addEventListener("click", function () {
    runs = runs + 1;
    show("Quick single!");
  });
  document.getElementById("four").addEventListener("click", function () {
    runs = runs + 4;
    show("Cracking four!");
  });
  document.getElementById("six").addEventListener("click", function () {
    runs = runs + 6;
    show("Into the stands!");
```

```
  });
  document.getElementById("out").addEventListener("click", function () {
    wickets = wickets + 1;
    show("Howzat! Out!");
    document.body.style.backgroundColor = "#fdecea";
  });
</script>
```

6. Debug it: the silent scoreboard (probe · web lab)

Goals checked: Finds the score by the right id · Listens for the right event

One solution:

```
<p id="score">Score: 0</p>
<button id="plus">+1</button>

<script>
  let score = 0;
  const scoreText = document.getElementById("score");
  const plus = document.getElementById("plus");
  plus.addEventListener("click", function () {
    score = score + 1;
    scoreText.textContent = "Score: " + score;
  });
</script>
```

Answer key — worksheet

1. **Key words:** JavaScript = The language that makes web pages react and move. · script tag = The tag that holds JavaScript code. · id = A unique name for one element, used to find it. · event = Something that happens, like a click.
2. **Plan your page:** Goals: Has a script; Finds elements by id; Listens for a click; Changes the words.
3. **Predict the score:** A. Runs: 11
4. **Match each piece of JavaScript to its job:** <script> → Holds the JavaScript code · let score = 0; → Makes a variable that starts at zero · document.getElementById("btn") → Finds the element with id="btn" · addEventListener("click", ...) → Waits for a click, then runs code · textContent → The words inside an element · score = score + 1; → Adds one to the score
5. **Order it: build a counter:** 1 Add a button and a paragraph, each with an id · 2 Add a script tag at the bottom of the page · 3 Make a variable: let clicks = 0 · 4 Find the button and paragraph with getElementById · 5 Listen for a click and add 1 to clicks · 6 Show the new number with textContent

Kitabash original curriculum.